Off-shell Supersymmetry and the M-theory Effective Action

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Based on work with
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Open string effective action

- Fradkin and Tseytlin (1985): Leading low-energy effective action of open string theory is Born-Infeld. Tseytlin's attempt at open superstring gives a weird answer.
- Cecotti and Ferrara (1986): 4D, $N = 1$ supersymmetry implies

$$\mathcal{L}_{CF} = \int d^2\theta W^2 + \int d^4\theta \, \Psi W^2 \bar{W}^2$$

  They found a $\Psi = \Psi(D^2 W^2, \bar{D}^2 \bar{W}^2)$ that reproduced Tseytlin's result but also observed that there is another corresponding to Born-Infeld again.
- Metsaev, Rahkmanov, and Tseytin realize the mistake (R boundary condition) and recalculate with NS to find that the low-energy effective action is Born-Infeld (1987).
- Bagger and Galperin demonstrate that, starting in 4D, $N = 1$ superspace, a second non-linearly-realized supersymmetry uniquely fixes the action to be Born-Infeld (1996). Schematically,

$$F^2 + F^4 + 8 \text{ supercharges} \Rightarrow \text{Born-Infeld}$$

Closed string effective action?

Can we carry out the analogous steps for closed strings/gravity?

- Cannot do Tseytin *et al.* calculation because world-sheet theory is interacting (quartic) even for constant Riemann tensor.
- Unclear what the gravitational analog of Born-Infeld is. Constant field strength? No birefringence? No ghosts? Determinantal form? Higher dimensions? Supersymmetric? . . . ?
- Analog of BG result may hold: There are only two supersymmetric densities in flat space. They would be related by $N \geq 2$ supersymmetry. Maybe even fixed by N large enough?

M-theory effective action?

- 11D SG has maximal D and N.
- First higher-derivative correction is of order (Riemann)^4.
- Higher-derivative correction requires a higher-derivative correction to the supersymmetry transformation, which requires higher derivatives in the effective action, *ad infinitum*.
  It is believed that in this case the analog of the Born-Infeld statement holds:

$$R + R^4 + 11\text{D},\ N = 1 \text{ supersymmetry} \Rightarrow \text{Leading M-theory LEEA?}$$

- Supersymmetrization order-by-order in components is unfeasible. Off-shell supersymmetry might fix this, but supersymmetry is on shell if $\#Q > 9$ (Berkovits 1993: Non-associativity of $\mathbb{O}$).
- Perhaps keeping some $0 < N < 1$ manifest in 11D is enough?

Previous work on fractional superspaces

- Marcus, Sagnotti, and Siegel (1983): 10D SYM in $\mathbf{R}^{10|4}$. The vector multiplet in this 10D, $N = 1/4$ superspace consists of a real field $V(x, \theta, \bar{\theta}; y)$ familiar from 4D, $N = 1$ and an $SU(3)$ triplet of chiral superfields $\Phi_i(x, \theta; y)$. Action $\sim \int d^{10}x \int d^2\theta W^\alpha W_\alpha +$ Chern-Simons terms.
- Berkovits (1994): Hybrid string in this superspace and open string field theory using the result of MSS (1995).
- Seiberg and Witten (1994): Starting with 4D, $N = 2$ SYM in terms of $N = 1$ superfields, there is a D-term integral of the Kähler function $K(\bar{\Phi}, \Phi)$ and an F-term integral of the gauge kinetic term determined by a holomorphic gauge kinetic function $f(\Phi)$. The second non-manifest supersymmetry determines the former in terms of the latter: $K = -\frac{i}{2} f' \bar{\Phi} + \text{h.c.}$
  Imposing $N = 4$ fixes $f(\Phi)$ uniquely.
- 5D, $N = 1/2$ linearized supergravity was constructed with Luty and Phillips (2002) and used to compute loop corrections to slepton masses in a phenomenological analog of heterotic M-theory (with also Buchbinder, Gates, Goh, and Ng).
- Extended to 11D SG with group at Texas A&M.

<u>Goal:</u> Higher-derivative corrections to the M-theory effective action.

Constructing 11D, $N = 1/8$ Superspace

We construct 11D supergravity in an off-shell superspace $\boldsymbol{M} \sim \boldsymbol{X} \times Y$ with $\boldsymbol{X}$ a 4D, $N = 1$ supermanifold and Y a bosonic 7D manifold.
This proceeds in six steps:

1. Decompose 11D SG fields

$$
\begin{array}{ccccc}
e_{\boldsymbol{m}}{}^{\boldsymbol{a}} & \to & e_m{}^a \ , \ \mathcal{A}^i_m \ , \ g_{ij} & \text{with} & a, m = 0, 1, 2, 3 \\
\psi^{\boldsymbol{\alpha}}_{\boldsymbol{m}} & \to & \psi^\alpha_m \ , \ \psi^{\alpha i}_m \ , \ \psi^\alpha_j \ , \ \psi^{\alpha i}_j \ , & \text{and} & i, j = 1, \ldots, 7 \\
C_{\boldsymbol{mnp}} & \to & C_{mnp} \ , \ C_{mn\,i} \ , \ C_{m\,ij} \ , \ C_{ijk} & \text{and} & \alpha = 1, 2
\end{array}
$$

Step 2 (of 6): Reinterpreting the Metric on Y

2 Rewrite the metric g_{ij} on Y in terms of a 3-form (Bryant 1987, Hitchin, Joyce 2000)

$$\sqrt{g}g_{ij} = \frac{1}{144}\epsilon^{k_1\cdots k_7}\varphi_{ik_1k_2}\varphi_{jk_3k_4}\varphi_{k_5k_6k_7}$$

- This 3-form is "stable" when g_{ij} is invertible. A stable 3-form on a 7 manifold is a(n almost) G_2 structure.
- The torsion of this G_2 structure is defined by $\nabla_l\varphi_{ijk} = T_l{}^m(*\varphi)_{ijkm}$.
- If $T_i^j = 0$, the holonomy of ∇ is contained in G_2. The $49 = 1 + 7 + 14 + 27$ components of this tensor make up the torsion forms

$$d\varphi = \tau_0 * \varphi + 3\tau_1 \wedge \varphi + *\tau_3 \quad \text{and} \quad d(*\varphi) = 4\tau_1 \wedge *\varphi + \tau_2 \wedge \varphi$$

- If, furthermore, Y is compact and simply-connected, then $\mathrm{Hol}(\nabla) = G_2$.
- The functional $\int_Y \sqrt{g(\varphi)} = \int_Y \varphi \wedge *\varphi$ is called the Hitchin (2000) functional. When $d\varphi = 0$ its equation of motion is $d(*\varphi) = 0 \Leftrightarrow G_2$ holonomy metrics ("Topological M-theory" Dijkgraaf, Gukov, Neitzke, Vafa 2004).

Step 3 (of 6): Superfield Embedding

3 Embed the bosonic fields into 11D, $N = 1/8$ superfields:

- ${e_m}^a$ and ψ_m^α in a real 4-vector-valued superfield $U^a(x, \theta, \bar{\theta}; y)$ familiar from (conformal) 4D, $N = 1$ supergravity,
- $\psi_m^{\alpha i}$ into a spinor superfield $\Psi_{\alpha i}(x, \theta, \bar{\theta}; y)$ in the $\bar{\mathbf{7}}$ of $GL(7)$,
- $\mathcal{A}_m^i$ into a real superfield $\mathcal{V}^i(x, \theta, \bar{\theta}; y)$ forming a vector on Y,
- C_{mnp} into a real scalar superfield $X(x, \theta, \bar{\theta}; y)$ forming a 0-form on Y,
- $C_{mn\,i}$ into a chiral spinor $\Sigma_{\alpha i}(x, \theta; y)$ forming a 1-form on Y,
- $C_{m\,ij}$ into 21 real superfields $V_{ij}(x, \theta, \bar{\theta}; y)$ forming a 2-form on Y,
- $C_{ijk} + i\varphi_{ijk}$ into 35 chiral superfields $\Phi_{ijk}(x, \theta; y)$ forming a 3-form on Y.

All these fields are superconformal primaries of the 4D, $N = 1$ superconformal algebra.

Step 4 (of 6): Chern-Simons Action

4 The last five form a "non-abelian tensor hierarchy" of superfields.[1] Such things have a unique Chern-Simons-like invariant supersymmetrizing the term $\int C \wedge dC \wedge dC$:

$$-12\kappa^2 \mathcal{L}_{CS} = i \int d^2\theta \mathcal{E} \left[i\Phi_{[0,3]} \wedge \mathbb{G}_{[4,4]} + \Sigma^{\alpha}_{[2,1]} \wedge \mathbb{W}_{[2,6]\alpha} \right] + \int d^4\theta E \left[V_{[1,2]} \wedge \mathbb{H}_{[3,5]} - X_{[3,0]} \mathbb{F}_{[1,7]} \right] + \text{h.c.}$$

- $\Phi \wedge \mathbb{G} = G\Phi \wedge d\Phi + \cdots$ contains "G_2 superpotential" (Gukov 1999).
- $\Phi \wedge \mathbb{G} = \frac{1}{2}\Phi \wedge W^{\alpha} \wedge W_{\alpha} + \cdots$ also contains the holomorphic gauge-kinetic function $f(\Phi) = \Phi$ familiar from Seiberg-Witten theory.

These terms and the others (10 total) are required for gauge invariance. Unique invariant containing F-term integrals: $\nexists$ gauge-invariant superpotential or gauge-kinetic function.

[1] Mathematically, this is a *Diff*(7)-equivariant chain complex of differential forms on Y with values in a complex of (sheaves of) representations of the 4D, $N = 1$ super-Poincaré algebra.

Steps 5 and 6: Complete Action

5 There is a natural lift of the Hitchin functional to 11D, $N = 1/8$ superspace:

$$S_H = -\frac{3}{\kappa^2}\int d^7y \int d^4x \int d^4\theta E(\bar{G}G)^{1/3}\sqrt{g(F)}\, f(x) \quad , \quad x = \frac{H_i g^{ij}(F) H_j}{(\bar{G}G)^{2/3}}$$

Uniquely fixed by local supersymmetry on $\boldsymbol{X}$, diffeomorphism invariance on Y, tensor hierarchy gauge invariance, engineering dimensions, conformal weights, $U(1)_R$ weights, and $GL(7)$ representation.

6 Introducing the gravitino superfield, we can extend the local supersymmetry on $\boldsymbol{X}$ to all of $\boldsymbol{M}$. This fixes $f(x)$ exactly:
The combination $\hat{f} := f - 2xf'$ satisfies the quartic equation

$$\frac{x}{4}\hat{f}^4 + \hat{f}^3 - 1 = 0 \quad \text{and} \quad f(0) = 1 \quad \Rightarrow \quad f(x)$$

Checks

We have subjected this result to the following consistency checks:

- Linearizing around a flat background, projecting to components, and throwing out fermions, we recover the correct bosonic component action of eleven-dimensional supergravity.
- Taking $\boldsymbol{M} = \mathbf{R}^{4|4} \times Y$ for general Y, reducing to components, throwing out all fermions and those bosons with polarizations in 4D directions, we obtain the "scalar potential"

$$V = \tfrac{1}{4} d_X^2 + 2 g_{ij} \boldsymbol{d}^i \boldsymbol{d}^j - d_{\mathbf{7}ij}^2 + \tfrac{1}{2} d_{\mathbf{14}ij}^2 - \tfrac{1}{18} |f_{\mathbf{1}ijk} + \tfrac{i}{2} d_X F_{ijk}|^2 - \tfrac{1}{24} |f_{\mathbf{7}ijk}|^2 + \tfrac{1}{24} |f_{\mathbf{27}ijk}|^2$$

  D- and F-term supersymmetry conditions $\Leftrightarrow$ G_2-holonomy and zero flux.

- Substituting the on-shell values of the auxiliary fields, this becomes

$$V(x,y) = -\tfrac{21}{16}\tau_0^2 - 15\tau_1^2 + \tfrac{1}{8}\tau_2^2 + \tfrac{1}{24}\tau_3^2 + \tfrac{7}{4}\sigma_0^2 + 9\sigma_1^2 + \tfrac{1}{24}\sigma_3^2$$

  G_2 identities for torsion forms and C-field analogs $\Rightarrow$

$$V(x,y) = -\tfrac{1}{2} R(g) + \tfrac{1}{4 \cdot 4!} F_{ijkl}^2$$

  $=$ Einstein-Hilbert and Maxwell terms on Y from *auxiliary field mechanism*.

Higher derivatives?

Can we guess higher-derivative terms? Many are easy. Most interesting one:

- We want to covariantize the term $\int C \wedge X_8$.
- It contains $\int C \wedge R \wedge R \wedge R \wedge R$.
- Decomposing, it contains $\int C \wedge W_X \wedge W_X \wedge W_Y \wedge W_Y$ (Weyl tensors of X, Y).
- $W_X \Leftrightarrow$ gravitino curl $W_{\alpha\beta\gamma}$ from 4D, $N = 1$ supergravity. So term $\Leftrightarrow$ $\varepsilon^{ijklmnp} \int \Phi_{ijk} W^{\alpha\beta\gamma} W_{\alpha\beta\gamma} S_{lmnp}$ for some $S \sim (W_Y)^2$.
- But the Riemann tensor on Y is real, so S cannot be constructed directly from it. This was also the case for the potential, so perhaps it is generated by some auxiliary field mechanism?
- But the Weyl tensor is in the $\mathbf{70'}$ representation of G_2, and there is no auxiliary field for this!

Strategies

We are pursuing a few strategies to study this problem.

1. Consider a simpler problem:
   - Construct Born-Infeld in 10D, $N = 1/4$ superspace.
   - Auxiliary fields in $SU(3)$ reps:
     - $\bar{\mathbf{3}} \leftrightarrow$ chiral $\Phi_i \leftrightarrow (2,0)$ part F_{ij} of the curvature of the connection in a Calabi-Yau compactification.
     - a singlet $\leftrightarrow$ vector field $V \leftrightarrow$ trace of $(1,1)$ part ${F_k}^k$.

     F-term SUSY $\Leftrightarrow$ holomorphic connex.
     D-term SUSY $\Leftrightarrow$ Donaldson-Uhlenbeck-Yau equation.
   - The analog of the $\mathbf{70'}$ above is the $\mathbf{8} = \binom{6}{2} - 3 - 3 - 1 \leftrightarrow {F_i}^j - \frac{1}{3}\delta_i{}^j {F_k}^k$ (the rest).

   Solution to the $\mathbf{8}$ problem in BI $\overset{?}{\longleftrightarrow}$ solution to $\mathbf{70'}$ problem in M-theory.
   Better yet: If we can show there is no solution to the $\mathbf{8}$ problem, we can safely forget about the $\mathbf{70'}$ problem and go work on something else.

Strategies

2nd Strategy: Shut up and calculate

1. Construct off-shell supergeometry
2. Construct $R \wedge R$ super-4-form
3. Square to get $R \wedge R \wedge R \wedge R$

To see how this works, first do easier 5D, $N = 1/2$ superspace:

1. Construct non-linear 5D, $N = 1/2$ superspace supergravity *à la* 11D, $N = 1/8$. (Linearized limit agrees 2003 result.)
2. We then use the known supergeometry (with Gates and Phillips 2003) to construct $R \wedge R$ (cf. Hanaki, Ohashi, and Tachikawa 2006):

$$\mathbb{G} = W^{\alpha\beta\gamma} W_{\alpha\beta\gamma}$$

$$\begin{aligned}\mathbb{H}_i = &-\tfrac{1}{4} D^\gamma F_i'^{\alpha\beta} W_{\alpha\beta\gamma} + \tfrac{1}{8} F_i'^{\alpha\beta} \bar{D}^{\dot\alpha} D_\beta G_{\underline{a}} + \tfrac{1}{8} \bar{D}^{\dot\alpha} F_i'^{\alpha\beta} D_\beta G_{\underline{a}} \\ &+ \tfrac{1}{16} \left(D^\beta \bar{D}_{\dot\alpha} F'_{\alpha\beta i} + \tfrac{1}{4} D_\alpha \bar{D}^2 \bar{\lambda}_{\dot\alpha i} \right) G^{\underline{a}} + \tfrac{1}{32} D^2 \lambda_i^\alpha D_\alpha R + \text{h.c.}\end{aligned}$$

3. This step is trivial in 5D.

Conveniently, this answer can be lifted to 11D, $N = 1/8$ unchanged.

Summary

- We have embedded eleven-dimensional supergravity in an off-shell 11D, $N = 1/8$ superspace.
- We have constructed the linearized off-shell supergeometry for it by starting with the fields and their gauge and local supersymmetry transformations and constructing the Riemann tensor, gravitino curls, and super-4-form field strength.
- We have partially constructed the $(\text{Riemann})^2$ invariants with the intention of using them to construct the leading $\sim R^4$ M-theory corrections.
- Along the way, we have revisited the 5D, $N = 1/2$ analog of this problem and are currently studying the 10D Born-Infeld analog.

Outlook

- Success in the construction of even the most non-trivial part of the $\int C \wedge X_8$ term is not the end of the story:
  1. Chern-Simons-like action is not invariant under all 11D super-diffeomorphisms (cf. 2-derivative action).
  2. D-term integral that fixes this problem already determined by $f(x)$. The non-invariance patched up by shift $G \to G + R \wedge R$ of tensor hierarchy field strengths (Witten 1996).
  3. But D-term integrand is non-polynomial $\Rightarrow$ non-polynomial dependence in Riemann.

  Outcome expected from general arguments:

  Supersymmetrization of R^4 must be non-polynomial.

Outlook

This would be an 11D supergravity analog of Born-Infeld. If this is correct, it would settle many questions about the definition of the latter but also leave some open and suggest new ones:

- It is not constructed purely from the curvature. Some derivatives thereof will be generated by the superspace integration. These are needed for supersymmetry.
- What is the fate of the singular GR solutions? This should not be difficult to answer once the action is known.
- Can the bosonic part be put in determinantal form? Does it define a new characteristic class?
- Is it the leading low-energy effective action of M-theory? This could be checked by comparing terms available in the literature from *e.g.* amplitude computations.
- Is there a 64-supercharge theory for which the M-theory result is the Goldstone? Witten's 12D theory and E_8 framing (1996)?
- ...

In addition, there are many applications/extensions (superstrings, compactifications, SFT, ...).

THANK YOU!

NATH fields and transformations

$$
\begin{aligned}
\delta_0 \Psi_{\alpha i} &= \Xi_{\alpha i} + \bar{G}\, \mathcal{G}_{ij} \nabla_\alpha \Omega^j + i\, W_{\alpha ij}\, \bar{\Omega}^j \ , \\
\Delta_1 \Phi &= \Xi^\alpha \wedge W_\alpha + \bar{\nabla}^2 \Big(\tfrac{i}{2} \iota_{\bar{\Omega}} \mathcal{U} + \tfrac{1}{4} \iota_{\bar{\Omega}} (F \wedge H) \Big) \ , \\
\Delta_1 V &= -\tfrac{i}{2} \bar{G} \iota_\Omega F + \tfrac{i}{2} G \iota_{\bar{\Omega}} F \ , \\
\Delta_1 \Sigma_\alpha &= -G\, \Xi_\alpha \ , \\
\Delta_1 X &= -\tfrac{i}{2} \bar{G}\, \iota_\Omega H + \tfrac{i}{2} G\, \iota_{\bar{\Omega}} H \ , \\
\delta_1 \mathcal{V} &= -\tfrac{1}{2} \bar{G}\, \Omega - \tfrac{1}{2} G\, \bar{\Omega} \ .
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{U}_{ijkl} &:= 3\, \epsilon_{ijklmnp} \frac{\partial}{\partial F_{mnp}} \Big(\sqrt{g}\, f\, (G\bar{G})^{1/3} \Big) \\
\mathcal{G}_{ij} &= (G\bar{G})^{-1/3} \Big(\hat{f}^{-1} g_{ij} + \tfrac{i}{2} (G\bar{G})^{-1/3} \hat{f} F_{ijk}\, g^{kl}\, H_l \Big)
\end{aligned}
$$

$$
\begin{aligned}
\delta_0 \Psi_{\alpha i} &= \Xi_{\alpha i} + g_{ij} D_\alpha \Omega^j \ , \\
\delta_1 \Phi_{ijk} &= \tfrac{1}{2i} \tilde{\varphi}_{ijkl} \bar{D}^2 \bar{\Omega}^l \ , \\
\delta_1 V_{ij} &= \tfrac{1}{2i} \varphi_{ijk} (\Omega^k - \bar{\Omega}^k) \ , \\
\delta_1 \Sigma_{\alpha i} &= -\Xi_{\alpha i} \ , \\
\delta_1 \mathcal{V}^i &= -\tfrac{1}{2} (\Omega^i + \bar{\Omega}^i) \ .
\end{aligned}
$$

NATH field strengths and Bianchi identities

$$
\begin{aligned}
E &= \partial\Phi \\
F &= \tfrac{1}{2i}\left(\Phi - \bar{\Phi}\right) - \partial V \\
W_\alpha &= -\tfrac{1}{4}\bar{\nabla}^2\nabla_\alpha V + \partial\Sigma_\alpha + \iota_{\mathcal{W}_\alpha}\Phi \\
H &= \tfrac{1}{2i}\left(\nabla^\alpha\Sigma_\alpha - \bar{\nabla}_{\dot{\alpha}}\bar{\Sigma}^{\dot{\alpha}}\right) - \partial X - \omega_{\mathsf{h}}(\mathcal{W}, V) \\
G &= -\tfrac{1}{4}\bar{\nabla}^2 X + \iota_{\mathcal{W}^\alpha}\Sigma_\alpha
\end{aligned}
$$

$$
\begin{aligned}
0 &= -\partial E \\
\tfrac{1}{2i}\left(E - \bar{E}\right) &= \partial F \\
-\tfrac{1}{4}\bar{\nabla}^2\nabla_\alpha F &= -\partial W_\alpha - \iota_{\mathcal{W}_\alpha} E \\
\tfrac{1}{2i}\left(\nabla^\alpha W_\alpha - \bar{\nabla}_{\dot{\alpha}}\bar{W}^{\dot{\alpha}}\right) &= \partial H + \omega_{\mathsf{h}}(\mathcal{W}, F) \\
-\tfrac{1}{4}\bar{\nabla}^2 H &= -\partial G - \iota_{\mathcal{W}^\alpha} W_\alpha \\
\bar{\nabla}_{\dot{\alpha}} G &= 0
\end{aligned}
$$

$$
\omega_{\mathsf{h}}(\chi_\alpha, v) := \iota_{\chi^\alpha}\nabla_\alpha v + \iota_{\bar{\chi}_{\dot{\alpha}}}\bar{\nabla}^{\dot{\alpha}} v + \tfrac{1}{2}\left(\iota_{\nabla^\alpha\chi_\alpha} v + \iota_{\bar{\nabla}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}} v\right)
$$

Chern-Simons-like invariant

$$
\begin{aligned}
-12L_{CS} = i \int d^2\theta\, \mathcal{E}\, \Phi \left[EG + \tfrac{1}{2} W^\alpha W_\alpha - \tfrac{i}{4} \bar{\nabla}^2 (FH) \right] \\
+ \int d^4\theta\, E\, V \left[\left(E + \bar{E} \right) H + \omega(W, F) - i \nabla^\alpha F \iota_{\mathcal{W}_\alpha} F + i \bar{\nabla}_{\dot{\alpha}} F \iota_{\bar{\mathcal{W}}^{\dot{\alpha}}} F \right] \\
+ i \int d^2\theta\, \mathcal{E}\, \Sigma^\alpha \left[E W_\alpha - \tfrac{i}{4} \bar{\nabla}^2 (F \nabla_\alpha F) \right] \\
- \int d^4\theta\, E\, X \left[\left(E + \bar{E} \right) F \right] + \text{h.c.}
\end{aligned}
$$